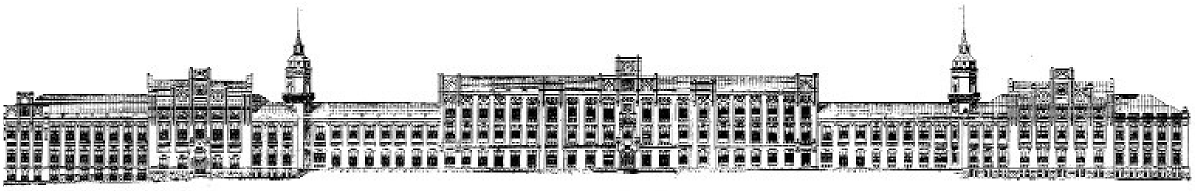

Міністерство освіти і науки України
Національний технічний університет України
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ЛАБОРАТОРНА РОБОТА №6

З дисципліни «Інфраструктура Інформаційних Технологій»

Тема: «ІНФРАСТРУКТУРА ЯК КОД. TERRAFORM»

Виконала:
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Перевірив:
Цимбал С. І.

2026

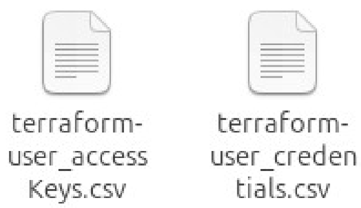
Тема: Інфраструктура як код. TERRAFORM.

Мета роботи: Ознайомитися з підходом інфраструктура як код. Опанувати використання інструмента Terraform для конфігурування, розгортання та управління інфраструктури в хмарі.

Хід роботи

1. Створити ключ-файл, за допомогою якого будете підключатися до інфраструктури (можна використати з минулого практикуму).

2. Створити IAM ключ, за допомогою якого буде доступ до створення інстансів, та зберегти ID та KEY згідно з інструкцією, з яким ви ознайомились під час опрацювання матеріалу.



3. Створити main.tf файл, в якому буде описано всю інфраструктуру застосунку (створення інстансу та Security group з минулого практикуму). Використовуйте репозиторій з минулого практикуму для розміщення в ньому необхідних файлів.

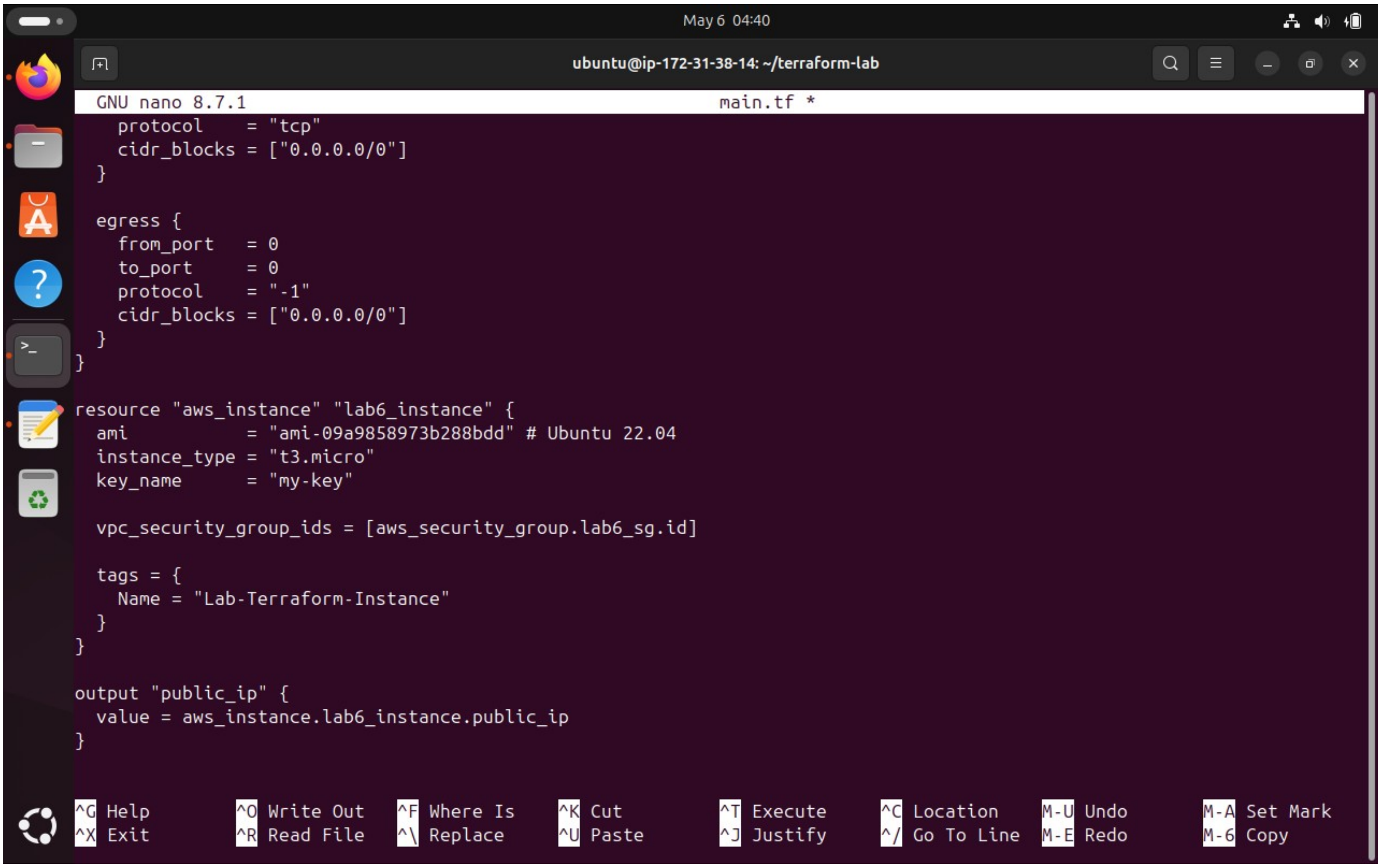
ubuntu@ip-172-31-35-129: ~

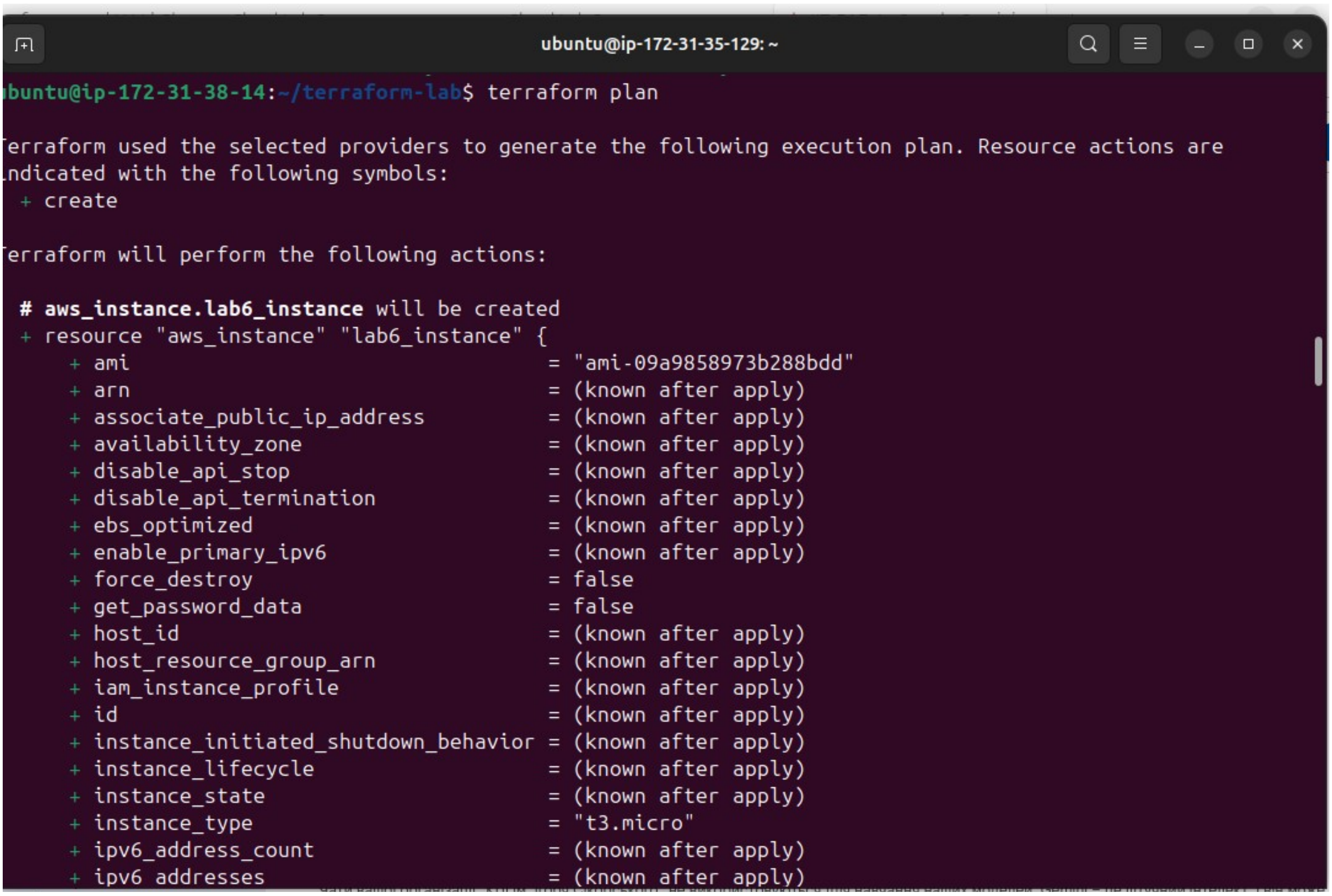
```
ubuntu@ip-172-31-38-14:~$ export AWS_ACCESS_KEY_ID="AKIAY2RLAAC7KOGHNFRG"
ubuntu@ip-172-31-38-14:~$ export AWS_SECRET_ACCESS_KEY="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"
ubuntu@ip-172-31-38-14:~$ export AWS_DEFAULT_REGION="eu-north-1"
ubuntu@ip-172-31-38-14:~$ mkdir ~/terraform-lab && cd ~/terraform-lab
ubuntu@ip-172-31-38-14:~/terraform-lab$ nano main.tf
ubuntu@ip-172-31-38-14:~/terraform-lab$ terraform init
Command 'terraform' not found, but can be installed with:
sudo snap install terraform
ubuntu@ip-172-31-38-14:~/terraform-lab$ sudo snap install terraform --classic
terraform 1.15.0 from Snapcrafters✪ installed
ubuntu@ip-172-31-38-14:~/terraform-lab$ terraform -version
Terraform v1.15.0
on linux_amd64

Your version of Terraform is out of date! The latest version
is 1.15.1. You can update by downloading from https://developer.hashicorp.com/terraform/install
ubuntu@ip-172-31-38-14:~/terraform-lab$ terraform init
Initializing provider plugins found in the configuration...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.43.0...
- Installed hashicorp/aws v6.43.0 (signed by HashiCorp)

Initializing the backend...


Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.
```

May 6 04:39

ubuntu@ip-172-31-38-14: ~/terraform-lab

```
  GNU nano 8.7.1                               main.tf *
provider "aws" {
  region = "eu-north-1"
}

resource "aws_security_group" "lab6_sg" {
  name        = "lab6-sg-ronyasha"
  description = "Allow SSH and HTTP"

  ingress {
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    from_port   = 80
    to_port     = 80
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }

^G Help       ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```


May 6 04:40
ubuntu@ip-172-31-38-14: ~/terraform-lab
GNU nano 8.7.1
main.tf *
    protocol   = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
  egress {
    from_port   = 0
    to_port   = 0
    protocol   = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
}
resource "aws_instance" "lab6_instance" {
  ami   = "ami-09a9858973b288bdd" # Ubuntu 22.04
  instance_type = "t3.micro"
  key_name   = "my-key"
  vpc_security_group_ids = [aws_security_group.lab6_sg.id]
  tags = {
    Name = "Lab-Terraform-Instance"
  }
}
output "public_ip" {
  value = aws_instance.lab6_instance.public_ip
}
^G Help
^O Write Out
^F Where Is
^K Cut
^T Execute
^C Location
M-U Undo
M-A Set Mark
^X Exit
^R Read File
^\ Replace
^U Paste
^J Justify
^/ Go To Line
M-E Redo
M-6 Copy


ubuntu@ip-172-31-35-129: ~
ubuntu@ip-172-31-38-14:~/terraform-lab$ terraform plan
erraform used the selected providers to generate the following execution plan. Resource actions are
ndicated with the following symbols:
  + create
erraform will perform the following actions:
  # aws_instance.lab6_instance will be created
  + resource "aws_instance" "lab6_instance" {
      + ami   = "ami-09a9858973b288bdd"
      + arn   = (known after apply)
      + associate_public_ip_address   = (known after apply)
      + availability_zone   = (known after apply)
      + disable_api_stop   = (known after apply)
      + disable_api_termination   = (known after apply)
      + ebs_optimized   = (known after apply)
      + enable_primary_ipv6   = (known after apply)
      + force_destroy   = false
      + get_password_data   = false
      + host_id   = (known after apply)
      + host_resource_group_arn   = (known after apply)
      + iam_instance_profile   = (known after apply)
      + id   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle   = (known after apply)
      + instance_state   = (known after apply)
      + instance_type   = "t3.micro"
      + ipv6_address_count   = (known after apply)
      + ipv6_addresses   = (known after apply)

ubuntu@ip-172-31-35-129: ~

```
Changes to Outputs:
  + public_ip = (known after apply)

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Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these
actions if you run "terraform apply" now.
ubuntu@ip-172-31-38-14:~/terraform-lab$ terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are
indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.lab6_instance will be created
  + resource "aws_instance" "lab6_instance" {
      + ami                                  = "ami-09a9858973b288bdd"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
```
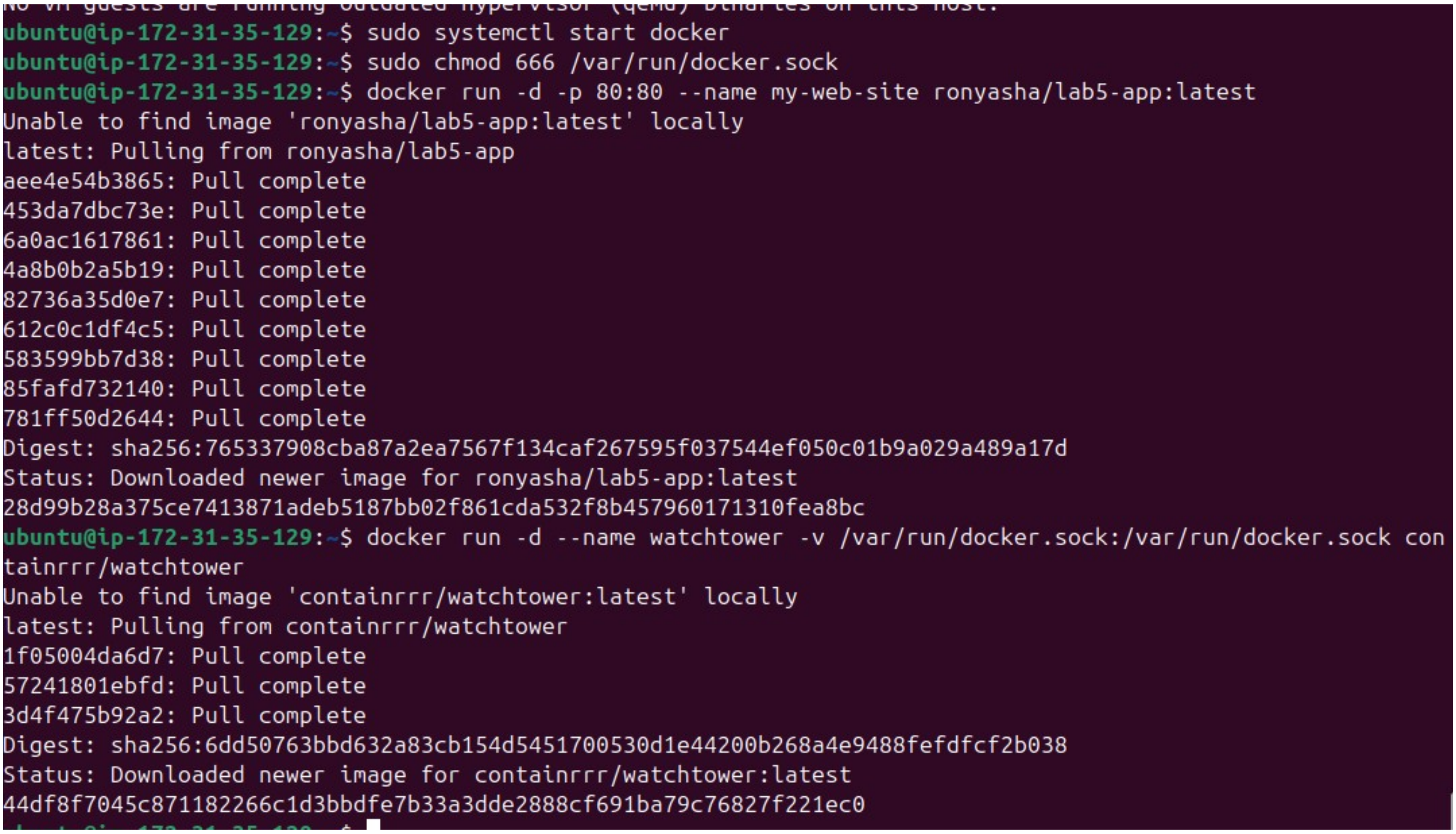
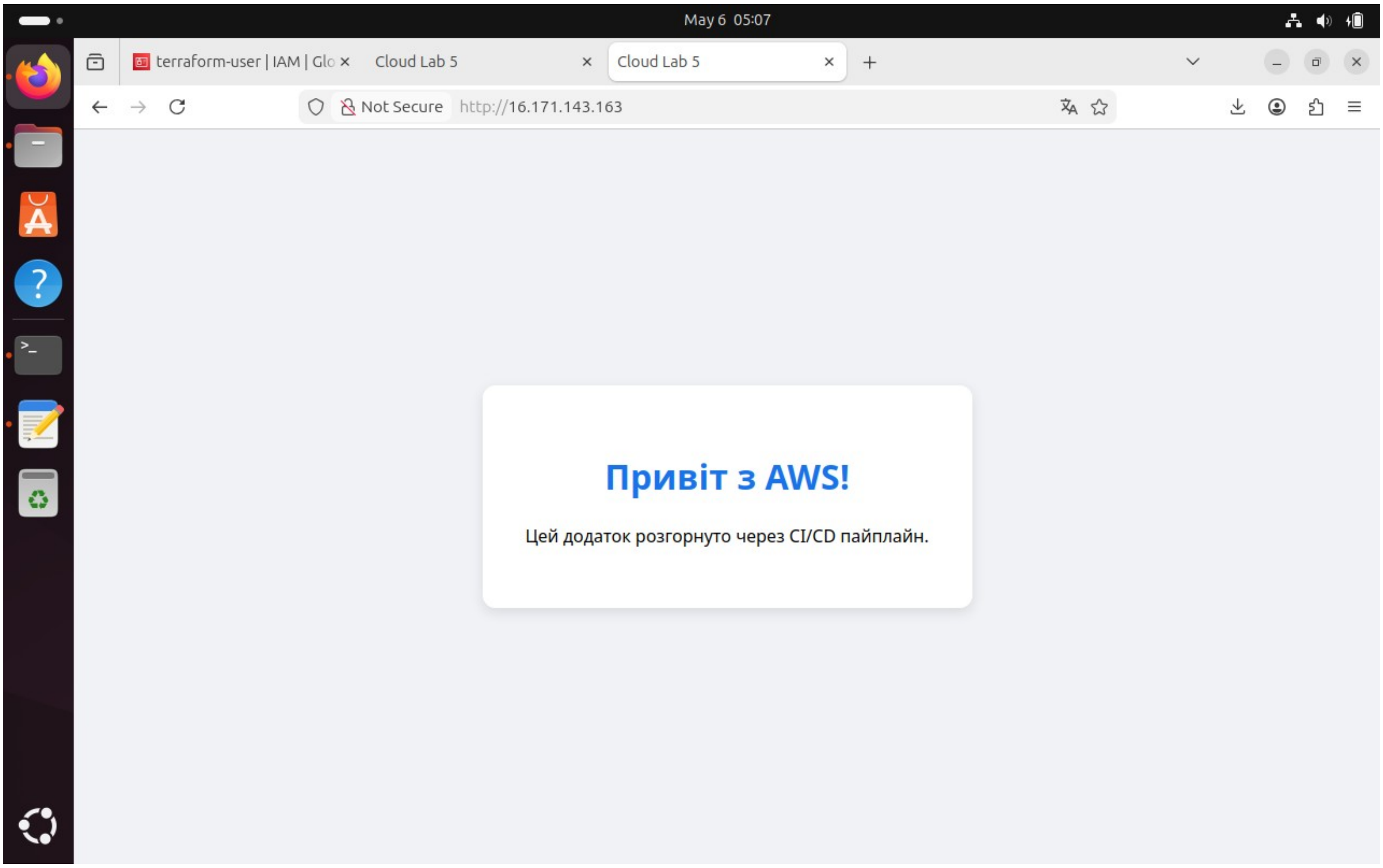

```
Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

public_ip = "16.171.143.163"
```

```
veronika@veronika-VirtualBox:~/Downloads$ cd ~/Downloads
veronika@veronika-VirtualBox:~/Downloads$ chmod 400 my-key.pem
veronika@veronika-VirtualBox:~/Downloads$ ssh -i my-key.pem ubuntu@16.171.143.163
The authenticity of host '16.171.143.163 (16.171.143.163)' can't be established.
ED25519 key fingerprint is SHA256:cGL3vkKbXTEedMhNzv+ISJVFuT1fF3B6ygzwyldgf2E.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '16.171.143.163' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1021-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed May  6 02:55:08 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   24.9% of 6.71GB   Processes:             107
  Memory usage: 23%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.35.129

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```


ubuntu@ip-172-31-35-129:~$ sudo systemctl start docker
ubuntu@ip-172-31-35-129:~$ sudo chmod 666 /var/run/docker.sock
ubuntu@ip-172-31-35-129:~$ docker run -d -p 80:80 --name my-web-site ronyasha/lab5-app:latest
Unable to find image 'ronyasha/lab5-app:latest' locally
latest: Pulling from ronyasha/lab5-app
aee4e54b3865: Pull complete
453da7dbc73e: Pull complete
6a0ac1617861: Pull complete
4a8b0b2a5b19: Pull complete
82736a35d0e7: Pull complete
612c0c1df4c5: Pull complete
583599bb7d38: Pull complete
85fafd732140: Pull complete
781ff50d2644: Pull complete
Digest: sha256:765337908cba87a2ea7567f134caf267595f037544ef050c01b9a029a489a17d
Status: Downloaded newer image for ronyasha/lab5-app:latest
28d99b28a375ce7413871adeb5187bb02f861cda532f8b457960171310fea8bc
ubuntu@ip-172-31-35-129:~$ docker run -d --name watchtower -v /var/run/docker.sock:/var/run/docker.sock con
tainrrr/watchtower
Unable to find image 'containrrr/watchtower:latest' locally
latest: Pulling from containrrr/watchtower
1f05004da6d7: Pull complete
57241801ebfd: Pull complete
3d4f475b92a2: Pull complete
Digest: sha256:6dd50763bbd632a83cb154d5451700530d1e44200b268a4e9488fefdfcf2b038
Status: Downloaded newer image for containrrr/watchtower:latest
44df8f7045c871182266c1d3bbdfe7b33a3dde2888cf691ba79c76827f221ec0


May 6 05:07
terraform-user | IAM | Glo
Cloud Lab 5
Cloud Lab 5
Not Secure
http://16.171.143.163
Привіт з AWS!
Цей додаток розгорнуто через CI/CD пайплайн.

Висновок: У ході виконання лабораторної роботи було опановано практичні навички роботи з інструментом Terraform для реалізації концепції Infrastructure as Code (IaC). Виконана робота продемонструвала переваги IaC, такі як швидкість розгортання, повторюваність та можливість версіонування інфраструктури разом із кодом застосунку.